

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

Course Code: CSE 219 Course Title: Signals and Linear Systems

Time: 25 minutes**Marks: 20**

Student Name: _____

Student ID: _____

Answer **exactly 2 out of Questions 1–4. Question 5 is mandatory.** Answer concisely.

1. Consider the signal $x[n]$ defined as follows for $N = 4$: [8]

$$x[n] = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } 1 \leq n < N \end{cases}$$

- Compute the 4-point radix-2 DIT FFT of the signal. Draw a butterfly diagram and show your calculations on the diagram.
 - If $x[n]$ is viewed as samples of an aperiodic continuous-time signal, what might be a continuous-time signal corresponding to it?
 - Based on your answer in (b), interpret your result from (a). If the sequence is zero-padded to length 32, what is its 32-point DFT? Briefly justify.
 - In a radix-2 DIT FFT, we typically use bit-reversed order for the input $x[n]$. Imagine you forgot to use bit-reversed order and used natural order for the input instead, resulting in a sequence $X'[k]$. Determine the signal $x'[n]$ whose DFT is actually $X'[k]$.
2. In many cases, the radix-4 FFT is highly efficient. Suppose you are deriving the radix-4 DIT Cooley-Tukey algorithm. You have already established that for $0 \leq k < N/4$: [8]

$$X[k] = G_0[k] + W_N^k G_1[k] + W_N^{2k} G_2[k] + W_N^{3k} G_3[k]$$

Where

$$G_m[k] = \sum_{r=0}^{N/4-1} x[4r+m] W_{N/4}^{kr}, \quad m = 0, 1, 2, 3$$

- Derive the formulas for $X[k + N/4]$, $X[k + 2N/4]$, and $X[k + 3N/4]$.

- Draw the radix-4 butterfly for the case $N = 4$.

Hint: Instead of just ± 1 , there will also be multiplications involving $\pm j$.

3. Consider two polynomials: [8]

$$A(x) = \sum_{n=0}^{N-1} a[n]x^n \quad \text{and} \quad B(x) = \sum_{n=0}^{N-1} b[n]x^n$$

You want to compute their product, $C(x) = A(x)B(x)$.

- Provide a simple, naive algorithm to find the polynomial $C(x)$ and analyze its time complexity.
 - If we express $C(x)$ as the sum $\sum c[n]x^n$, determine the valid range for n and provide a general formula for the coefficients $c[n]$.
 - Express the coefficient sequence $c[n]$ as a convolution.
 - Propose a more efficient algorithm to compute $C(x)$ and analyze the time complexity of your proposed algorithm.
4. In image processing, we often use a 2-dimensional version of the DFT, which is defined as: [8]

$$X[k_1, k_2] = \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} x[n_1, n_2] W_{N_1}^{k_1 n_1} W_{N_2}^{k_2 n_2}$$

Where $x[n_1, n_2]$ is an $N_1 \times N_2$ matrix and $W_N^k = e^{-j2\pi k/N}$.

- What is the time complexity of a naive algorithm used to compute the 2D DFT?
 - How can you interpret the given formula as two successive 1D DFT operations?
 - Propose an efficient algorithm (2D FFT) for computing this and analyze its complexity.
 - What is the primary difference between your 2D FFT algorithm and Bailey's FFT algorithm?
5. Thank you for attending the classes and participating in this CT. [4]